NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Foot & Ankle

1. flatfoot and cavus foot
2. hallux valgus and forefoot
3. Achilles and tendons

Flatfoot and cavus foot

Part A: Adult acquired flatfoot (progressive collapsing foot deformity)

1. Terminology and anatomy

- **Adult acquired flatfoot deformity (AAFD)**, renamed **progressive collapsing foot deformity (PCFD)** (2020 consensus): **complex 3D deformity** with **hindfoot valgus, forefoot abduction (talonavicular uncoverage), forefoot varus (supination), medial column instability (collapse of the arch)**, and later **ankle valgus**.
- **Tibialis posterior (TP): strongest invertor; primary dynamic stabiliser of the medial longitudinal arch**; inserts mainly on the **navicular tuberosity** with slips to the cuneiforms, cuboid and metatarsal bases 2-4; **locks the transverse tarsal (Chopart) joint** (TN and CC axes become non-parallel: rigid lever for push-off); **antagonist: peroneus brevis**.
- **Hypovascular zone** of TP tendon: **behind and just distal to the medial malleolus** (degeneration site).
- **Static stabilisers: spring ligament (calcaneonavicular: superomedial part most important)**, **deltoid ligament (deep)**, **plantar fascia (windlass mechanism)**, **talocalcaneal interosseous ligament**.
- **Posterior tibial tendon dysfunction (PTTD)** was the classic cause, but **spring ligament failure** is now recognised as central; TP failure is often secondary.

2. Causes

- **TP tendon degeneration (commonest in middle-aged obese women; hypertension, diabetes, steroid injections, seronegative arthropathy)**.
- **Others: inflammatory arthritis (RA)**, **trauma (TP laceration/rupture, Lisfranc injury)**, **neuropathic (Charcot)**, **tarsal coalition (adult**

presentation), **accessory navicular**, **degenerative TMT/naviculocuneiform arthritis**, **iatrogenic**.

3. Clinical evaluation

- **Medial ankle pain along the TP tendon** (early), **lateral sinus tarsi/subfibular impingement pain** (late: calcaneofibular impingement).
- **"Too many toes" sign** (**forefoot abduction seen from behind**).
- **Single-limb heel rise test**: **inability to perform / no heel inversion on heel rise** (**most sensitive test for TP dysfunction**); **double heel rise** to check flexibility (hindfoot inverts if flexible).
- **Flexibility of the hindfoot (subtalar)** and **forefoot (forefoot varus/supination: compensatory fixed or flexible)**: **assess forefoot position with the hindfoot held in neutral**.
- **Gastrocnemius/Achilles contracture** (**Silfverskiöld test**: almost universal).
- **Medial column instability** (first TMT/naviculocuneiform plantar sag), **deltoid laxity (ankle valgus)**.

4. Imaging

- **Weight-bearing radiographs (AP and lateral foot, AP ankle, hindfoot alignment view (Saltzman))**:
  - **Lateral talo-first metatarsal angle (Meary angle): normal 0 degrees (collinear)**; **over 4 degrees convex plantar = flatfoot** (**mild 4-15, moderate 15-30, severe over 30**),
  - **Calcaneal pitch angle (normal about 20-30 degrees; reduced in flatfoot)**,
  - **AP talonavicular coverage angle (over 7 degrees) / talonavicular uncoverage percentage (over 40%)**: forefoot abduction,
  - **AP talo-first metatarsal angle** (**abduction**),
  - **Medial cuneiform-floor height** (reduced), **sinus tarsi impingement**, **subfibular impingement**.
  - **Ankle: talar tilt (valgus) in the mortise** (deltoid failure).

- **Weight-bearing CT** (**3D, subfibular impingement, middle facet subluxation**: **peritalar subluxation**).
- **MRI** (TP tendon tears/tendinosis, spring ligament), **ultrasound**.

5. Classification

Johnson and Strom (modified by Myerson/Bluman)

Stage	TP tendon	Hindfoot	Treatment
I	**Tenosynovitis/tendinosis; normal length**	**Normal alignment; can single heel rise** (painful)	**Non-operative (immobilisation, NSAIDs, orthoses)**; **tenosynovectomy**
II (A/B/C)	**Elongated/dysfunctional**	**Flexible flatfoot** (**IIA mild abduction; IIB abduction over 30-40% TN uncoverage; IIC fixed forefoot varus**)	**Joint-sparing reconstruction**: **FDL transfer to the navicular + medializing calcaneal osteotomy (MDCO) ± lateral column lengthening (Evans) ± Cotton osteotomy ± gastrocnemius recession ± spring ligament repair**
III	**Ruptured/degenerated**	**Rigid flatfoot (fixed hindfoot valgus)**	**Triple arthrodesis** (or **double/modified: TN + subtalar**)
IV (A/B)		**+ ankle valgus (deltoid failure)**: **IVA flexible (correctable), IVB rigid (tibiotalar arthritis)**	**IVA: deltoid reconstruction + hindfoot correction**; **IVB: tibiotalocalcaneal (TTC) or pantalar arthrodesis, or TAR + hindfoot correction**

PCFD consensus classification (2020)

- **Stage 1 (flexible) or 2 (rigid)**, with **classes A (hindfoot valgus), B (midfoot/forefoot abduction), C (forefoot varus/medial column instability), D (peritalar subluxation/dislocation), E (ankle instability: valgus)**; each scored as flexible or rigid.

6. Treatment details

- **Non-operative (all stages first, mainstay in elderly/low demand): CAM boot or short leg cast (acute tenosynovitis, 4-8 weeks)**, **orthoses (medial arch support + medial heel wedge; UCBL)**, **ankle-foot orthosis (Arizona brace)** for later stages, **physiotherapy (eccentric TP strengthening, Achilles stretching)**, weight loss. **Avoid steroid injection into the TP tendon sheath (rupture)**.
- **Flexible deformity (stage II) procedures:**
  - **FDL transfer to the navicular** (**FDL is in phase with and adjacent to the TP; weaker (about 30%) but adequate when combined with bony correction**; **FHL** alternative; **interconnection (Henry knot) allows the FHL to flex lesser toes**),
  - **Medializing displacement calcaneal osteotomy (MDCO)** (**about 10 mm medial shift**: **converts the Achilles tendon pull into an invertor (medial to the subtalar axis)**; **protects the tendon transfer**; **sural nerve and lateral calcaneal branch at risk; the tibial nerve/tarsal tunnel at risk with large shifts**),
  - **Lateral column lengthening (Evans osteotomy: anterior calcaneus between the anterior and middle facets, about 10-15 mm from the CC joint; or CC distraction arthrodesis)** for **forefoot abduction (TN uncoverage over about 40%)**: **lateral column overload (lateral foot pain: commonest complication)**, **CC arthritis**, **sural nerve injury**, non-union,
  - **Cotton osteotomy (plantarflexion opening-wedge osteotomy of the medial cuneiform)** or **first TMT arthrodesis (Lapidus)** for **residual forefoot varus/medial column instability**,
  - **Spring ligament repair/reconstruction**,
  - **Gastrocnemius recession (Strayer) / Achilles lengthening (Hoke)** for equinus,

  - **Subtalar arthroereisis** (sinus tarsi implant: limited role in adults; mainly paediatric).
- **Rigid deformity (stage III): triple arthrodesis (subtalar, TN, CC)** through **medial and lateral incisions (or single medial incision: protects lateral skin)**; **TN joint is the key joint (fusion of TN alone eliminates almost all hindfoot motion)**; **position: hindfoot about 5 degrees valgus, neutral forefoot rotation (plantigrade), neutral abduction**; **complications: non-union (TN commonest site), malunion, adjacent joint arthritis (ankle, naviculocuneiform)**.
- **Stage IV: deltoid ligament reconstruction** (flexible) or **TTC/pantalar arthrodesis** (rigid with arthritis); **total ankle replacement with hindfoot realignment**.

Part B: Pes cavus

7. Definition and causes

- **Elevated medial longitudinal arch (high arch), often with hindfoot varus**; **forefoot-driven (plantarflexed first ray) or hindfoot-driven**.
- **About two-thirds have an underlying neurological cause**: **Charcot-Marie-Tooth disease (HMSN type I: commonest; autosomal dominant; PMP22 duplication on chromosome 17p)**, **spinal cord pathology (tethered cord, diastematomyelia, syringomyelia, tumour, spinal dysraphism)**, **cerebral palsy**, **poliomyelitis (India: residual polio)**, **Friedreich ataxia**, **stroke**, **muscular dystrophy**. **Also: idiopathic/familial (subtle cavovarus)**, **residual clubfoot**, **trauma (compartment syndrome: Volkmann's ischaemic contracture: claw toes; malunited talar neck fracture)**.
- **Unilateral cavus = spinal cord lesion until proved otherwise: MRI whole spine. Bilateral cavus: think CMT (nerve conduction studies, genetic testing)**.

8. CMT pathomechanics

- **Muscle imbalance (pairs of agonist/antagonist; weakest muscles first: tibialis anterior and peroneus brevis; intrinsics):**
  - **Weak tibialis anterior vs strong peroneus longus**: **plantarflexed first ray (forefoot pronation/valgus)**,
  - **Weak peroneus brevis vs strong tibialis posterior**: **hindfoot varus, forefoot adduction**,
  - **Intrinsic weakness**: **claw toes** (**EDL recruited to dorsiflex the ankle: hyperextends MTPJs**),
  - **Plantar fascia contracture** (windlass: elevates the arch).
- **Tripod effect**: **plantarflexed first ray forces the hindfoot into varus on weight bearing (forefoot-driven hindfoot varus)**.
- **Lateral ankle instability, peroneal tendon tears, fifth metatarsal stress fractures, lateral foot callosities, stress fractures, ankle arthritis (varus)**.

9. Evaluation

- **Peek-a-boo heel sign** (**medial heel pad visible from the front with the feet parallel: hindfoot varus**).
- **Coleman block test** (**block under the lateral foot (heel and lateral forefoot), allowing the first ray to drop off the edge**): **hindfoot varus corrects = flexible hindfoot, deformity is forefoot-driven** (treat the forefoot: **first metatarsal dorsiflexion osteotomy + plantar fascia release**); **does not correct = fixed hindfoot varus (needs calcaneal osteotomy or arthrodesis)**.
- **Muscle power testing** (MRC grading of each muscle: guides tendon transfers), **neurological examination** (reflexes, sensation, spine), **calluses (under first and fifth metatarsal heads)**.
- **Radiographs: Meary angle (talo-first metatarsal) convex dorsally (negative: over -4 degrees = cavus)**, **calcaneal pitch angle increased (over 30 degrees: calcaneocavus)**, **Hibbs angle** (calcaneus axis vs first metatarsal axis: **under 150 degrees** = cavus), **"double talar dome"** (ankle AP/lateral: hindfoot varus/external rotation), **fibula posterior**.

10. Treatment

- **Non-operative: orthoses (lateral forefoot post + recess under the first metatarsal head)**, **lateral heel wedge** (for flexible hindfoot), **AFO** (foot drop), **physiotherapy**, **footwear**.
- **Operative (individualised; "flexible deformities need soft-tissue/tendon procedures, fixed deformities need bony procedures, arthrodesis for severe rigid deformity or arthritis"):**
  - **Soft tissue: plantar fascia release (Steindler)**, **gastrocnemius recession/Achilles lengthening** (**in calcaneocavus (polio/spina bifida) the Achilles is weak: do not lengthen**).
  - **Tendon transfers (balance forces): peroneus longus to peroneus brevis transfer** (**removes plantarflexion of the first ray; augments eversion**), **tibialis posterior transfer to the dorsum (through the interosseous membrane to the lateral cuneiform)** for **foot drop**, **Jones transfer (EHL to the first metatarsal neck + hallux IPJ fusion)** for **clawed hallux**, **Hibbs transfer (EDL to the lateral cuneiform/cuboid)** for **claw lesser toes**, **flexor-to-extensor transfer (Girdlestone-Taylor)** for flexible claw toes.
  - **Bony (joint-sparing): dorsiflexion closing-wedge osteotomy of the first metatarsal base** (**forefoot-driven cavus**), **lateral closing-wedge calcaneal osteotomy (Dwyer)** or **lateralising calcaneal (Z) osteotomy** (**fixed hindfoot varus**), **midfoot osteotomy (Cole: dorsal closing wedge through the navicular-cuneiform/cuboid; Japas V-osteotomy; Akron dome)** for **midfoot cavus**, **anterior tibial osteotomy** rarely.
  - **Arthrodesis: triple arthrodesis** (**rigid deformity, arthritis, salvage**; **worse long-term results in CMT (ankle/midfoot arthritis)**: **avoid when possible in young patients**).
  - **Lateral ankle ligament reconstruction** (Broström) **must be combined with correction of the varus** (otherwise fails).

11. Paediatric flatfoot (brief link)

- **Flexible flatfoot: physiological; no treatment unless painful + tight heel cord (calcaneal lengthening + gastrocnemius recession)**.
- **Rigid flatfoot: tarsal coalition, CVT, accessory navicular, inflammatory, neuromuscular**: see paediatric foot notes.

12. Other conditions

- **Tibialis anterior tendon rupture: elderly men (sixth-seventh decade)**, **degenerative**, **foot slap/steppage gait**, **palpable gap/mass at the anterior ankle**, **EHL/EDL substitution (toe clawing on dorsiflexion)**; **repair/reconstruction (EHL transfer, graft) in active patients**; **AFO in low-demand**.
- **Accessory navicular (os naviculare)** in adults: **Kidner procedure (excision + TP advancement)**.
- **Tarsal tunnel syndrome: tibial nerve compression beneath the flexor retinaculum (laciniate ligament)**; **space-occupying lesions (ganglion, varicosities, lipoma, accessory muscles: flexor digitorum accessorius longus)**, **hindfoot valgus (traction)**, trauma; **burning plantar paraesthesia (sparing the heel (calcaneal branch arises proximal) in some cases), Tinel sign, dorsiflexion-eversion test**; **NCS**; **MRI**; **decompression (release of the flexor retinaculum + abductor hallucis fascia (medial and lateral plantar nerve tunnels))**: **results less predictable than CTS (about 50-75%)**, best with a **space-occupying lesion**.
- **Baxter nerve (first branch of the lateral plantar nerve / inferior calcaneal nerve) entrapment**: **chronic heel pain (medial)** mimicking plantar fasciitis; **abductor digiti minimi atrophy on MRI**; decompression with plantar fascia release.
- **Plantar fasciitis (heel pain): degenerative fasciosis at the medial calcaneal tubercle origin**; **first-step morning pain (post-static dyskinesia)**; **calcaneal spur (in FDB origin) is incidental**; **non-operative (about 90%: plantar fascia-specific stretching (best), Achilles stretching, heel cups, night splints, NSAIDs)**; **steroid**

injection (risk of fat pad atrophy and plantar fascia rupture), **ESWT**; **surgery (partial plantar fascia release (medial third to half: avoid complete release: lateral column overload)** ± **Baxter nerve release, gastrocnemius recession)** after **6-12 months**.

Exam pearl: **PCFD/AAFD: TP + spring ligament failure; too many toes; can't single-heel-rise (most sensitive). Johnson-Strom: I tendinosis (conservative), II flexible (FDL transfer + MDCO ± Evans ± Cotton ± gastroc recession), III rigid (triple arthrodesis), IV ankle valgus (deltoid recon or TTC/TAR). MDCO converts Achilles to an invertor; Evans = lateral column overload.**

Exam pearl: **Meary angle: plantar convex = flatfoot, dorsal convex = cavus. Calcaneal pitch high = calcaneocavus. TN joint key to triple arthrodesis (TN non-union commonest); position 5 degrees valgus.**

Exam pearl: **Cavus: unilateral = spinal MRI; bilateral = CMT (PMP22). CMT: weak TA vs strong PL (plantarflexed first ray), weak PB vs strong TP (varus). Coleman block: corrects = forefoot-driven (first MT dorsiflexion osteotomy + plantar fascia release + PL to PB); does not correct = add Dwyer/lateralising calcaneal osteotomy. Jones transfer for clawed hallux. Broström fails without varus correction.**

Exam pearl: **Tarsal tunnel: flexor retinaculum, space-occupying lesion gives best release results. Plantar fasciitis: stretching best; partial release only. Baxter nerve: ADM atrophy.**

Hallux valgus and forefoot

Part A: Hallux valgus

1. Definition and aetiology

- **Lateral deviation of the hallux at the first MTPJ with medial deviation of the first metatarsal (metatarsus primus varus)**, with a medial prominence ("bunion").
- **Women (about 10:1 in surgical series)**, **familial (about 60-90% of juvenile cases; maternal transmission)**, **constricting footwear (narrow toe box, high heels)**, **ligamentous laxity**, **pes planus**, **first ray hypermobility (debated)**, **long first metatarsal**, **rounded metatarsal head**, **increased DMAA**, **inflammatory arthritis (RA)**, **neuromuscular (cerebral palsy, stroke)**, **amputation of the second toe**.

2. Pathoanatomy (sequence)

1. **Medial capsule and medial sesamoid ligament attenuation**; **the hallux drifts laterally (valgus)** pulled by **adductor hallucis** and the lateral pull of the **EHL and FHL** (**bowstring effect: become deforming forces**).
2. **The metatarsal head drifts medially off the sesamoids** (the **sesamoids stay in place: anchored by the adductor hallucis and the transverse metatarsal ligament**): **apparent lateral sesamoid displacement** relative to the metatarsal head; **crista erosion**.
3. **Pronation (eversion) of the hallux** (abductor hallucis migrates plantarly: loses abduction power; becomes a flexor/pronator).
4. **Lateral capsule and adductor contracture**; **transfer metatarsalgia** under the second metatarsal head (first ray loses load-bearing).
5. **Second toe deformities** (crossover toe, hammer toe), **second MTPJ instability/plantar plate tear**.

3. Evaluation

- **Weight-bearing AP and lateral radiographs** (and sesamoid axial view).
- **Normal values and severity:**

Angle	Normal	Mild	Moderate	Severe
Hallux valgus angle (HVA: first MT axis vs proximal phalanx axis)	**Under 15 degrees**	**15-20** (up to 20-25 in some schemes)	**20-40**	**Over 40**
Intermetatarsal angle (IMA: 1st-2nd MT)	**Under 9 degrees**	**9-11** (under 13)	**11-16** (13-20)	**Over 16** (over 20)
Distal metatarsal articular angle (DMAA)	**Under 10 degrees** (about 6-10)			
Hallux valgus interphalangeus (HVI)	**Under 10 degrees**			

- **Congruent vs incongruent joint: congruent (articular surfaces parallel; increased DMAA; juvenile)** vs **incongruent (subluxed: lateral subluxation of the phalanx on the head)**. **Congruent deformity: correction must not be done by soft-tissue release alone (would create an incongruent joint): use DMAA-correcting osteotomy (biplanar chevron) or extra-articular (double/triple) osteotomies**.
- **Sesamoid position (Hardy and Clapham: grades 1-7)**, **first MTPJ arthritis** (**hallux rigidus: fusion instead**), **first TMT hypermobility** (**dorsal translation over about 8-10 mm: Lapidus**), **metatarsus adductus** (**increased difficulty; IMA underestimated**), **pronation of the first metatarsal (round lateral head sign)**.

4. Management

Non-operative

- **Wide toe-box footwear, bunion pads, orthoses, toe spacers**; **do not correct the deformity**; **surgery is for pain, not cosmesis** ("**never operate for cosmesis**").

Operative principles: choose by severity, joint congruency, arthritis and TMT stability

Deformity	Procedure	Notes
Mild (HVA under 30, IMA under 13), congruent or incongruent	**Distal metatarsal osteotomy: Chevron (Austin)** (**V-shaped osteotomy at the metatarsal head, apex distal; lateral shift up to about 30-50% of the head width**) ± **distal soft-tissue procedure (lateral release)**	**AVN of the metatarsal head** (**lateral release + chevron: avoid excessive dorsolateral dissection: nutrient vessels enter dorsolaterally (first dorsal metatarsal artery) and plantar-laterally**); **biplanar chevron for increased DMAA**
Moderate (HVA 30-40, IMA 13-20)	**Scarf osteotomy** (**Z-shaped diaphyseal osteotomy**: lateral translation; **troughing (fragments telescope into each other: elevation of the head, loss of height)**), **proximal metatarsal osteotomy (proximal chevron, crescentic, opening/closing wedge, Ludloff, Mau)** + **distal soft-tissue release** (modified McBride)	**Proximal osteotomies: more powerful correction (each degree of correction at the base produces more displacement distally)**; **dorsal malunion (elevatus: transfer metatarsalgia)**
Severe (HVA over 40, IMA over 20)	**Proximal osteotomy + distal soft-tissue procedure**, **Lapidus (first TMT arthrodesis)**	

Deformity	Procedure	Notes
First TMT hypermobility / generalised laxity / recurrent HV	**Modified Lapidus (first TMT arthrodesis)**	**Non-union (about 5-10%)**, **shortening (transfer metatarsalgia)**, dorsiflexion malunion
Arthritis of the first MTPJ, severe deformity, neuromuscular, RA, salvage of failed surgery	**First MTPJ arthrodesis**	**Position: about 10-15 degrees valgus, about 10-15 degrees dorsiflexion relative to the floor (about 20-25 degrees relative to the first metatarsal shaft), neutral rotation**; **IMA corrects spontaneously** after fusion
Hallux valgus interphalangeus	**Akin osteotomy** (**medial closing-wedge osteotomy of the proximal phalanx**)	Adjunct to other osteotomies
Elderly, low-demand	**Keller resection arthroplasty** (**excision of the base of the proximal phalanx**)	**Cock-up deformity, transfer metatarsalgia, weakness of push-off**; largely abandoned
Juvenile/adolescent	**Delay until skeletal maturity** (**high recurrence**); **guided growth (lateral hemiepiphysiodesis of the first metatarsal)**	**Increased DMAA common**

- **Simple bunionectomy (Silver) alone**: **high recurrence; not used alone**.
- **McBride procedure** (original): **excision of the lateral sesamoid** (**hallux varus risk**), **adductor tenotomy**, **medial capsulorrhaphy**: **modified McBride = lateral release without sesamoid excision**.

- **Minimally invasive (MIS) third/fourth generation (MICA/percutaneous chevron-Akin)**: increasingly used; similar outcomes; learning curve.

5. Complications

- **Recurrence (commonest; about 10-25%)**: **undercorrection of the IMA**, **unrecognised increased DMAA**, **unaddressed pronation**, **first TMT hypermobility**, **juvenile**, **poor sesamoid reduction**.
- **Hallux varus** (**overcorrection: excessive lateral release, lateral sesamoid excision, excessive medial eminence resection (staking the sagittal sulcus), overtightened medial capsule, IMA overcorrection (negative)**): **flexible: EHB tenodesis/EHL transfer; rigid/arthritic: MTPJ fusion**.
- **AVN of the metatarsal head** (**chevron + extensive lateral release**).
- **Transfer metatarsalgia** (**shortening or dorsal malunion of the first metatarsal**).
- **Stiffness of the first MTPJ**, **non-union/malunion**, **infection**, **dorsomedial cutaneous nerve injury (neuroma)** (**medial dorsal cutaneous branch of the superficial peroneal nerve**), **CRPS**.

Part B: Hallux rigidus

6. Hallux rigidus (first MTPJ OA)

- **Commonest arthritis of the foot**; **dorsal osteophyte (dorsal bunion)** limiting **dorsiflexion (pain at push-off)**; **positive grind test (late)**; **adolescent form (OCD of the metatarsal head)**.
- **Coughlin and Shurnas classification (grades 0-4): 0 stiffness, normal radiograph**; **1 mild, dorsal osteophyte, preserved joint space**; **2 moderate, joint space narrowing, pain at extremes**; **3 severe, under 10-20% residual motion, pain at extremes, more narrowing**; **4: as 3 with pain in mid-range (positive grind test)**.
- **Treatment:**

- **Non-operative: stiff-soled shoe, rocker sole, Morton's extension (carbon plate)**, NSAIDs, steroid injection.
- **Cheilectomy** (**excision of the dorsal osteophyte and dorsal 25-30% of the metatarsal head**): **grades 1-2 (and selected 3)**; **pain at the extremes, not mid-range**; **± Moberg osteotomy (dorsal closing-wedge osteotomy of the proximal phalanx: shifts the arc into dorsiflexion)**.
- **First MTPJ arthrodesis**: **gold standard for grades 3-4** (**predictable pain relief**; **position: 10-15 degrees dorsiflexion relative to the floor, 10-15 degrees valgus, neutral rotation**).
- **Interpositional arthroplasty/Keller** (elderly), **hemiarthroplasty (synthetic cartilage implant, Cartiva: mixed results)**; **silicone/total joint arthroplasty: high failure (silicone synovitis, subsidence)**.

Part C: Lesser toe deformities and forefoot pain

7. Lesser toe deformities

Deformity	MTPJ	PIPJ	DIPJ	Notes
Mallet toe	Neutral	Neutral	**Flexed**	**FDL contracture**
Hammer toe	**Neutral or slightly extended**	**Flexed**	**Neutral/extended**	**Commonest; second toe**; FDL/FDB contracture
Claw toe	**Hyperextended**	**Flexed**	**Flexed**	**Usually multiple toes, bilateral: neuromuscular (CMT, diabetes neuropathy), RA, cavus foot: intrinsic weakness** (lumbrical/interossei)

Deformity	MTPJ	PIPJ	DIPJ	Notes
Curly toe		Flexed + varus (rotated)		**Congenital** (FDL tightness)

- **Flexible deformities: flexor-to-extensor transfer (Girdlestone-Taylor: FDL split to the extensor hood/dorsum of the proximal phalanx)**, **FDL tenotomy (mallet)**.
- **Fixed deformities: PIPJ resection arthroplasty (DuVries) or PIPJ arthrodesis**; **MTPJ: extensor tendon lengthening (EDL Z-lengthening), EDB tenotomy, dorsal capsulotomy**; **Weil osteotomy** (**oblique shortening osteotomy of the metatarsal neck**: for **MTPJ dislocation/subluxation, long metatarsal**: **floating toe** = commonest complication).
- **Crossover second toe / second MTPJ instability (plantar plate tear): dorsomedial drift of the second toe**, **pain under the second MTPJ**, **positive drawer (Lachman) test of the MTPJ**; **Coughlin plantar plate grading**; **plantar plate repair + Weil osteotomy**, flexor-to-extensor transfer.
- **Freiberg disease** (second metatarsal head osteochondrosis: adolescent girls; dorsal closing-wedge osteotomy).

8. Metatarsalgia

- **Pain under the metatarsal heads**: **primary** (**long second metatarsal**, plantarflexed metatarsal, **cavus foot**, prominent condyles), **secondary** (**inflammatory arthritis, Freiberg, MTPJ synovitis, gastrocnemius tightness**), **iatrogenic (transfer after hallux valgus surgery)**.
- **Intractable plantar keratosis (IPK)**: **discrete** (beneath a prominent fibular condyle) vs **diffuse** (beneath a long metatarsal).
- **Gastrocnemius contracture (positive Silfverskiöld test: ankle dorsiflexion limited with the knee extended, improves with the knee flexed)**: **gastrocnemius recession (Strayer)**.
- **Treatment: metatarsal pad (placed proximal to the metatarsal heads)**, rocker-bottom sole, orthoses; **Weil osteotomy**, **condylectomy**.

9. Morton neuroma (interdigital neuroma)

- **Perineural fibrosis of the common digital nerve** (**not a true neuroma**), **distal to the deep transverse metatarsal ligament**; **third web space (between 3rd and 4th metatarsal heads) commonest**, then second; **women (about 5-10:1)**, **narrow/high-heeled shoes**.
- **Burning plantar forefoot pain radiating into the toes**, **numbness**, **"walking on a pebble"**; **Mulder click (squeezing the forefoot mediolaterally while pressing the web space)**, **web space tenderness**; **splaying of the toes (Sullivan sign)**.
- **Imaging: ultrasound or MRI** (size **over about 5 mm**); **differentials: MTPJ synovitis/plantar plate tear (second web), stress fracture, intermetatarsal bursitis, tarsal tunnel syndrome, lumbar radiculopathy**.
- **Treatment: wide shoes, metatarsal pad, steroid injection (up to about 50-80% relief)**, **alcohol sclerosing injections**, **radiofrequency ablation**; **surgery: neurectomy (dorsal approach preferred: avoids plantar scar; divide the deep transverse metatarsal ligament; resect the nerve proximal to the ligament, about 3 cm)** (**about 80-90% success**); **recurrent (stump) neuroma (about 5-10%)**: **plantar approach** for revision; **decompression (ligament release) alone** as an alternative.

10. Sesamoid disorders

- **Sesamoiditis, stress fracture, AVN, osteochondritis, OA, bipartite sesamoid (medial/tibial sesamoid commonest: about 10-30%; bilateral in about 85%: compare with the other foot)**, **turf toe**.
- **Turf toe: hyperdorsiflexion injury of the first MTPJ (plantar plate/capsuloligamentous complex)**: **American football on artificial turf**; **grade I** sprain, **II** partial tear, **III** complete tear (**proximal migration of sesamoids**, **sesamoid diastasis**, **vertical instability: Lachman positive**); **grade III: surgical repair** of the plantar plate.
- **Treatment of sesamoid pain:** offloading (dancer's pad, orthoses), boot/cast; **sesamoidectomy** (refractory: **excision of both sesamoids**

causes cock-up deformity: avoid; **medial sesamoid excision: hallux valgus**; **lateral sesamoid excision: hallux varus**).

11. Bunionette (tailor's bunion)

- **Lateral prominence of the fifth metatarsal head**; **Coughlin types: I enlarged/prominent head**, **II lateral bowing of the fifth metatarsal**, **III increased 4-5 intermetatarsal angle (over 8-10 degrees: commonest symptomatic)**.
- **Treatment: wide shoes**; **type I: lateral condylectomy** or **distal chevron osteotomy**; **type II: distal chevron or diaphyseal (oblique/scarf) osteotomy**; **type III: diaphyseal oblique or proximal osteotomy** (**proximal fifth metatarsal osteotomy: poor blood supply: non-union**).

Exam pearl: **HVA under 15 and IMA under 9 normal; DMAA under 10. Sesamoids stay; the metatarsal head drifts medially. Congruent joint (increased DMAA, juvenile) needs DMAA-correcting osteotomy, not soft-tissue release. Mild: chevron; moderate: scarf/proximal osteotomy; severe/hypermobile: Lapidus; arthritic: fusion (10-15 valgus, 10-15 dorsiflexion to floor). Commonest complication: recurrence; hallux varus = overcorrection (lateral sesamoid excision).**

Exam pearl: **Hallux rigidus: Coughlin-Shurnas; cheilectomy for pain at the extremes (grade 1-2, ± Moberg); mid-range pain/grind (3-4) = fusion.**

Exam pearl: **Claw toe (MTPJ hyperextension): neuromuscular/cavus; flexible: Girdlestone-Taylor (FDL to extensor). Crossover second toe = plantar plate tear (drawer test): repair + Weil (floating toe). Morton neuroma: 3rd web, Mulder click, distal to DTML; neurectomy via dorsal approach. Bunionette type III (increased 4-5 IMA) = diaphyseal osteotomy.**

Achilles and tendons

Part A: Achilles tendon

1. Anatomy

- **Formed by the gastrocnemius (medial and lateral heads: cross the knee) and soleus** (± plantaris medially); **largest and strongest tendon in the body** (loads up to **about 12 times body weight** in running).
- **Fibres rotate about 90 degrees** (medial fibres become posterior, **soleus fibres insert medially**) as they descend: stress concentration.
- **Paratenon** (no true synovial sheath), **insertion on the middle third of the posterior calcaneal tuberosity** (footprint); **retrocalcaneal bursa** (between tendon and calcaneus) and **retro-Achilles (subcutaneous) bursa**.
- **Blood supply: posterior tibial artery (proximal and distal)**, **peroneal artery (mid-portion)**; **hypovascular watershed zone about 2-6 cm proximal to the insertion** (**site of most ruptures and mid-portion tendinopathy**).
- **Sural nerve**: crosses from medial to lateral, **lateral to the tendon at about 10 cm proximal to the insertion** (**at risk in lateral incisions and percutaneous repairs**).

2. Acute Achilles tendon rupture

- **Epidemiology: men aged 30-50 ("weekend warriors")**, **sudden eccentric loading** (push-off with knee extended, unexpected dorsiflexion), **left more than right**; **2-6 cm above the insertion (commonest)**.
- **Risk factors: fluoroquinolones**, **corticosteroids (systemic or local injection)**, **pre-existing tendinopathy**, inflammatory arthritis, gout, **renal failure, hyperparathyroidism**, previous contralateral rupture.
- **Clinical: "kicked in the back of the leg"**, **audible pop**, **weak plantarflexion (but often preserved through FHL, FDL, TP, peronei)**:

missed in about 20%.
  - **Simmonds-Thompson test (calf squeeze: absent plantarflexion = rupture)** (**most reliable**),
  - **Palpable gap**,
  - **Matles test** (**loss of resting plantarflexion tension: the injured foot hangs in neutral/dorsiflexion with knees flexed prone**),
  - **Increased passive dorsiflexion**, **O'Brien needle test**, **Copeland sphygmomanometer test**.
  - **Two of the clinical tests (AAOS guideline) sufficient for diagnosis.**
- **Imaging: ultrasound** (gap, **gap distance in plantarflexion (over 5-10 mm)**: may guide treatment); **MRI** (chronic/equivocal); **radiograph: Kager's fat pad distortion (triangle obscured), avulsion fracture (calcaneal tuberosity avulsion: "beak" fracture: urgent surgery: skin necrosis)**.

Management

	Non-operative (functional rehabilitation)	Operative (open or minimally invasive)
Re-rupture	**About 3-12% (similar to surgery when early functional rehab is used)**	**About 2-5%**
Complications	Re-rupture; **tendon elongation (weaker push-off)**	**Wound complications/infection (about 5-10% open), sural nerve injury (percutaneous: about 5-10%)**, DVT
Strength	Possibly slightly weaker (elongation)	Slightly better plantarflexion strength in some studies

- **Evidence: early functional rehabilitation (early weight bearing in an equinus boot with heel wedges)** reduces the re-rupture difference: **Willits (2010)**, **UKSTAR trial (2020, UK: functional bracing vs plaster casting: similar outcomes)**, **Scandinavian trials (Nilsson-Helander,**

Ochen meta-analysis): **surgery reduces re-rupture absolute risk by about 2-3% at the cost of more complications**; **choice individualised** (**athletes, delayed presentation, large gap, avulsion: surgery**).

- **Non-operative protocol: equinus (about 20-30 degrees plantarflexion) in a boot with wedges**, **early protected weight bearing (2 weeks)**, **gradual reduction of equinus over 6-8 weeks**, **avoid passive dorsiflexion stretching for 12 weeks**; return to sport about 6-9 months.
- **Operative: open repair (Krackow/Bunnell core sutures + epitendinous; medial incision avoids the sural nerve)**, **minimally invasive/percutaneous (Achillon, PARS, Ma-Griffith)** (**fewer wound problems, sural nerve risk**); **repair with the ankle in resting tension (compare with the contralateral side)**.
- **Calcaneal tuberosity (sleeve) avulsion fracture: urgent ORIF (skin at risk)**.

3. Chronic (neglected) Achilles rupture

- **Over 4-6 weeks**; **gap with scar**; **weak push-off, inability to single heel rise**, **increased passive dorsiflexion**.
- **Kuwada classification** (gap): **I partial**, **II complete with gap under 3 cm (end-to-end repair)**, **III gap 3-6 cm (V-Y advancement / gastrocnemius recession / tendon graft)**, **IV gap over 6 cm (tendon transfer + graft)**.
- **Myerson: under 1-2 cm: end-to-end repair**; **2-5 cm: V-Y lengthening (gastrocnemius aponeurosis) ± FHL transfer**; **over 5 cm: FHL transfer (± turndown flap, allograft)**.
- **FHL transfer** (**strongest, in phase, adjacent, vascularised muscle belly brings blood supply, low donor morbidity**) to the calcaneus (**tunnel**); **FDL, peroneus brevis** alternatives; **gastrocnemius turndown flap (Bosworth/Lindholm)**.

4. Achilles tendinopathy

	Mid-portion (non-insertional)	Insertional
Location	**2-6 cm proximal to the insertion**	**At the calcaneal insertion**
Patients	**Runners, active middle-aged**	**Older, less active, overweight**, inflammatory arthritis (seronegative)
Pathology	**Tendinosis (degeneration, neovascularisation)**, **paratenonitis** (crepitus, fixed tender area)	**Calcification/enthesophyte (spur)**, **Haglund deformity (posterosuperior calcaneal prominence)**, **retrocalcaneal bursitis**
Signs	**Painful fusiform swelling that moves with the tendon (painful arc sign)**; **Royal London Hospital test**	**Posterior heel tenderness, bump ("pump bump")**
First-line	**Eccentric loading (Alfredson protocol: 3 x 15 reps, twice daily, 12 weeks, knee straight and bent)** or **heavy slow resistance**; **ESWT**; **high-volume injection**, sclerotherapy	**Eccentric loading on flat ground (not into dorsiflexion)**, **heel lift**, **shoe modification (open-backed)**, **ESWT**
Surgery (after 6 months)	**Debridement/longitudinal tenotomies**, **stripping of paratenon/neovessels**, **plantaris release (medial)**, **gastrocnemius recession**, **FHL transfer if over 50% of tendon debrided**	**Haglund resection + debridement of calcification + reattachment (suture anchors)** (**central tendon-splitting approach**); **FHL transfer** for extensive degeneration (over 50%); **Zadek (dorsal closing-wedge calcaneal) osteotomy**

- **Haglund syndrome: Haglund deformity + retrocalcaneal bursitis + insertional tendinopathy**; **radiographic: Fowler-Philip angle (over 75 degrees)**, **parallel pitch lines (Pavlov)**.

- **Avoid steroid injections into the Achilles** (rupture); **fluoroquinolones** caution.

Part B: Other tendons

5. Peroneal tendons

- **Anatomy: peroneus brevis (anterior/medial, closer to the fibula, in the retromalleolar groove; inserts on the base of the fifth metatarsal)**, **peroneus longus (posterior/lateral; crosses under the cuboid in the cuboid tunnel to insert on the base of the first metatarsal and medial cuneiform)**; **superior peroneal retinaculum (SPR: from the posterolateral fibula to the calcaneus)**; **fibrocartilaginous rim** deepens the groove; **os peroneum** (sesamoid in PL at the cuboid: about 20%).
- **Peroneus brevis split tears: longitudinal splits from compression against the fibula** (**cavovarus foot, ankle instability, low-lying PB muscle belly, peroneus quartus**): **lateral ankle pain/swelling**; **MRI ("boomerang/chevron" shape)**; **debridement and tubularisation (if under 50% involved)**, **tenodesis to PL (if over 50%)**; **correct cavovarus**.
- **Peroneal tendon subluxation/dislocation: forceful dorsiflexion with peroneal contraction (skiing)**; **SPR avulsion with a "fleck sign" (rim fracture of the lateral malleolus: Eckert-Davis grade 3)**; **Eckert-Davis/Oden grades (1: SPR stripped from the fibula with periosteum (commonest); 2: fibrocartilaginous rim elevated with SPR; 3: bony avulsion; 4: SPR torn from calcaneal attachment)**. **Acute: cast trial (about 50% success) or SPR repair**; **chronic: SPR repair/reconstruction + fibular groove deepening**. **Intrasheath subluxation** (tendons swap position within the groove).
- **Painful os peroneum syndrome (POPS): PL rupture at the cuboid notch/os peroneum fracture**; **proximal migration of the os peroneum on serial radiographs = PL rupture**; excision + PB tenodesis.

6. Tibialis posterior

- See flatfoot notes (PCFD).

7. Flexor hallucis longus

- **FHL tenosynovitis ("dancer's tendinitis")**: **ballet dancers (en pointe)**: **posteromedial ankle pain, triggering of the hallux ("hallux saltans")**, **pain on resisted hallux flexion**; **stenosis at the fibro-osseous tunnel behind the talus (between the medial and lateral talar tubercles)**; associated with **posterior ankle impingement (os trigonum/Stieda process)**: **rest, physiotherapy; release of the FHL tunnel ± os trigonum excision (open posteromedial or arthroscopic/endoscopic posterior)**.
- **Knot of Henry**: FHL-FDL crossover (plantar midfoot).

8. Tibialis anterior

- **Rupture: elderly men, degenerative**; see flatfoot notes.

9. Posterior ankle impingement

- **Os trigonum** (**unfused lateral tubercle of the posterior talar process: about 7-14%**) or **Stieda process** (elongated lateral tubercle); **ballet dancers, footballers (forced plantarflexion: nutcracker)**; **posterior ankle pain in plantarflexion (forced plantarflexion test)**, **diagnostic injection**; **excision (posterior arthroscopy: posterolateral and posteromedial portals with the patient prone; FHL is the landmark: stay lateral to FHL to protect the tibial neurovascular bundle)**.
- **Anterior ankle impingement: "footballer's ankle"**: **anterior tibial and talar neck osteophytes (kissing)** (**bony**) or **anterolateral soft-tissue impingement (Bassett ligament: accessory anteroinferior tibiofibular ligament fascicle; meniscoid lesion (Wolin))**; **arthroscopic debridement**; **anterior arthroscopy portals: anteromedial (medial to the tibialis anterior) and anterolateral (lateral to peroneus tertius: superficial peroneal nerve (intermediate dorsal cutaneous branch)**

at risk: commonest complication of ankle arthroscopy); **avoid the anterocentral portal (dorsalis pedis/deep peroneal nerve)**.

10. Plantar fascia rupture

- **Acute rupture** (often after steroid injection), **plantar heel pain and bruising**; non-operative (boot).

11. Gastrocnemius-soleus problems

- **Tennis leg** (**medial head of gastrocnemius tear at the musculotendinous junction**; middle-aged; "shot in the calf"); **differentiate from DVT (ultrasound)**; non-operative.
- **Isolated gastrocnemius contracture** (**Silfverskiöld test**): associated with **metatarsalgia, plantar fasciitis, Achilles tendinopathy, PCFD, midfoot arthritis**: **gastrocnemius recession (Strayer: distal aponeurosis; Baumann: intramuscular proximal)** (**sural nerve at risk**); **Achilles lengthening (Hoke triple hemisection: percutaneous)** when both gastrocnemius and soleus are tight (**overlengthening: calcaneal gait/weak push-off**).

> Exam pearl: **Achilles: watershed 2-6 cm above insertion; sural nerve lateral at about 10 cm. Thompson test most reliable. Early functional rehab narrows re-rupture difference (UKSTAR, Willits); surgery: fewer re-ruptures, more wound/sural complications. Chronic gap over 5 cm: FHL transfer (± V-Y/turndown).**

> Exam pearl: **Mid-portion tendinopathy: eccentric Alfredson (12 weeks), plantaris release. Insertional: no dorsiflexion eccentrics; Haglund resection + debridement + reattachment; FHL if over 50% debrided. Avoid steroid injections and fluoroquinolones.**

Exam pearl: **Peroneus brevis split tears (cavovarus): under 50% repair, over 50% tenodesis to PL. Peroneal dislocation: SPR, fleck sign; groove deepening for chronic. Os peroneum migration = PL rupture. Dancers: FHL tendinitis + os trigonum (stay lateral to FHL arthroscopically). Ankle arthroscopy: anterolateral portal risks the superficial peroneal nerve.**